

TECHNICAL NOTE

Phase-Specific Energy per Token for LLaMA-3 8B

Estimations for an NVIDIA A100 server and a OnePlus 13-class edge platform

September 1, 2026 | Values rounded to two significant digits

Platform / mode	Prefill J per input token	Decode J per output token	Evidence status
A100 80GB, batch 1	≈0.017	≈3.1	Cross-study estimate
A100, throughput-oriented serving	not isolated	≈0.18–0.27	Direct aggregate measurement
OnePlus 13-class, batch 1	≈0.013	≈0.28	Same-SoC transfer estimate

Table 1. Recommended phase-specific values. “OnePlus 13-class” is not a direct OnePlus 13 power measurement.

1. Method and Scope

The prefill phase processes all prompt tokens in parallel, whereas decode generates output tokens autoregressively. The phase-specific energy intensity is calculated from average phase power P and phase throughput R :

$$E_{\text{prefill}}/\text{token} = P_{\text{prefill}} / R_{\text{prefill}} \quad \text{and} \quad E_{\text{decode}}/\text{token} = P_{\text{decode}} / R_{\text{decode}}$$

The numerator and denominator must refer to the same phase, model, precision, backend, batching policy, and power scope. The estimates below retain those conditions where possible and explicitly label cross-study or cross-device transfers. They should not be interpreted as hardware constants.

2. NVIDIA A100 Server Estimate

Configuration. One NVIDIA A100-SXM4 80GB GPU, LLaMA-3 8B in FP16/BF16, SGLang, batch size 1, and a 4K context. WaferLLM reports 13,988 input tokens/s for prefill and 78 output tokens/s for decode on this baseline [1]. OASIS independently measures 239.4 W average GPU power for LLaMA-3 8B on an A100 at 1320 MHz [2].

Prefill estimate. Applying the measured A100 power to the reported prefill throughput gives:

$$239.4 \text{ W} / 13,988 \text{ input token s}^{-1} = 0.0171 \text{ J per input token}$$

Decode estimate. For an interactive single-request workload:

$$239.4 \text{ W} / 78 \text{ output token s}^{-1} = 3.07 \text{ J per output token}$$

The decode estimate is deliberately labeled cross-study: the throughput and power were not measured in the same run. A reasonable uncertainty interval is approximately 2–3.5 J/output token. Under high concurrency, the result changes substantially. OASIS directly reports 261.7 mJ/token at 1320 MHz and 171.5 mJ/token at 855 MHz, because GPU power is amortized across 847–915 aggregate tokens/s [2]. These aggregate serving values are not phase-isolated batch-one measurements.

3. OnePlus 13-Class Edge Estimate

Configuration. The OnePlus 13 contains a Snapdragon 8 Elite SoC [4]. The closest phase-resolved public evidence is HeteroInfer, which evaluates a LLaMA-8B/LLaMA-3.1-8B-class model using W4A16 weight-only quantization, batch size 1, and a GPU–NPU heterogeneous backend [3,5]. Most power results were obtained on Snapdragon 8 Gen 3; the paper separately measures Snapdragon 8 Elite performance. Therefore, the following values are transfer estimates rather than direct OnePlus 13 measurements.

Prefill. HeteroInfer reports 236.92 input tokens/s for a 256-token prompt on Snapdragon 8 Gen 3 and approximately 10.5% higher prefill performance on Snapdragon 8 Elite [3]. Thus:

$$\text{R8 Elite, prefill} \approx 236.92 \times 1.105 = 261.8 \text{ input tokens/s}$$

Digitizing the Hetero-tensor prefill power bar in Figure 20 gives approximately 3.3 W. Transferring that power to the Snapdragon 8 Elite throughput yields:

$$3.3 \text{ W} / 261.8 \text{ input token s}^{-1} = 0.0126 \text{ J per input token}$$

Decode. HeteroInfer reports 14.01 output tokens/s for LLaMA-8B and states that decode performance remains similar on Snapdragon 8 Elite because the tested platforms have similar effective memory bandwidth [3]. Figure 20 indicates approximately 3.9–4.0 W during heterogeneous decode. Therefore:

$$(3.9\text{--}4.0) \text{ W} / 14.01 \text{ output token s}^{-1} = 0.278\text{--}0.286 \text{ J per output token}$$

Recommended reporting ranges are 0.010–0.018 J/input token for prefill and 0.25–0.35 J/output token for decode. The ranges cover plot digitization, device-specific thermal behavior, and transferring Snapdragon 8 Gen 3 power to the Snapdragon 8 Elite-based OnePlus 13.

4. Interpretation and Reporting Guidance

For batch-one interactive inference, the A100 processes prompts dramatically faster, but its decode energy is dominated by low GPU utilization. The phone is much slower but draws only about 4 W, producing an estimated decode energy roughly one order of magnitude below a batch-one A100. With production batching, however, the A100 amortizes its power across many concurrent sequences and approaches 0.18–0.27 J/token—comparable to the mobile estimate. Consequently, any proposal comparison should state batch size, context length, precision, backend, and whether power covers the accelerator or the whole device.

Recommended proposal wording. For LLaMA-3 8B, we estimate approximately 0.017 J/input token and 3.1 J/output token on a batch-one A100, versus approximately 0.013 J/input token and 0.28 J/output token on a OnePlus 13-class Snapdragon 8 Elite platform. These are configuration-specific, literature-grounded estimates; the mobile values are not direct OnePlus 13 measurements, and batched A100 serving can reduce output-token energy to approximately 0.18–0.27 J/token.

References

- [1] He et al., “WaferLLM: Large Language Model Inference at Wafer Scale,” OSDI 2025, Tables 7–8. [Source](#)
- [2] Mittal et al., “OASIS: Optimal Allocation Strategy for Inference Services in Cloud Environments,” ICPE Companion 2026, Table 3. [Source](#)
- [3] Chen et al., “Characterizing Mobile SoC for Accelerating Heterogeneous LLM Inference,” SOSP 2025, Figures 16, 17, and 20. [Source](#)
- [4] OnePlus, “OnePlus 13 Specifications,” Snapdragon 8 Elite platform. [Source](#)
- [5] Qualcomm AI Hub, “Llama-v3.1-8B-Chat Quantized.” [Source](#)